

SMART TRAFFIC SIGNAL CONTROLLER USING STM32 Blue Pill

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Abstract—Rapid urbanization and the exponential growth in the number of vehicles have made traffic congestion one of the most critical challenges faced by modern cities, where conventional fixed-time traffic signal systems fail to respond effectively to real-time traffic variations, resulting in increased waiting time, inefficient traffic flow, higher fuel consumption, environmental pollution, and serious pedestrian safety concerns. To address these issues, this project proposes the design and implementation of a Smart Traffic Signal Controller using an STM32 microcontroller, which provides an adaptive, efficient, and intelligent traffic management solution. The system is developed using FreeRTOS to enable real-time multitasking and efficient handling of multiple operations at a four-way intersection. A structured time-based scheduling algorithm is used to control the sequence of traffic signals, while an advanced event-driven mechanism dynamically adjusts signal timings based on traffic density and priority conditions, ensuring that roads with higher vehicle load receive longer green signals. Additionally, the system incorporates a pedestrian request mechanism, which temporarily overrides normal traffic flow to ensure safe and controlled pedestrian crossing. For enhanced monitoring and data analysis, UART communication is employed to transmit real-time traffic data to an external interface, where a Python-based monitoring system logs the data into Excel files and sends live updates through Telegram for remote access and supervision. Furthermore, a prediction model is integrated to analyze traffic patterns by comparing actual and expected usage, detecting imbalances, and suggesting optimized adaptive signal timings for improved efficiency. By combining real-time control, intelligent decision-making, communication, and predictive analytics, the proposed system offers a comprehensive and scalable solution that significantly reduces traffic congestion, minimizes delays, improves fuel efficiency, enhances pedestrian safety, and supports the development of smart and sustainable urban transportation systems.

Index Terms—Smart Traffic Signal Controller, STM32 Microcontroller, FreeRTOS, Adaptive Traffic Control, Traffic Density Detection, Event-Driven System, Embedded Systems, Real-Time Monitoring, UART Communication, Python-Based Monitoring, Traffic Prediction, Intelligent Transportation System (ITS), Smart City Applications

I. INTRODUCTION

Rapid urbanization and the continuous growth in the number of vehicles have led to serious traffic congestion problems in modern cities. Most existing traffic signal systems operate on fixed-time intervals, which are unable to adapt to real-time traffic conditions. This results in inefficient traffic flow,

unnecessary waiting time, increased fuel consumption, and higher levels of air pollution. In addition, these systems do not effectively handle pedestrian crossing requests, leading to safety risks and inconvenience. Traffic conditions at intersections vary dynamically depending on factors such as vehicle density, peak hours, and road usage patterns. However, traditional systems lack the intelligence to respond to these variations, causing delays even when certain roads have minimal traffic. This highlights the need for a smart and adaptive traffic management system that can make decisions based on real-time data. To address these challenges, this project proposes a Smart Traffic Signal Controller using an STM32 microcontroller. The system is designed to dynamically adjust signal timings based on traffic density using a combination of time-based scheduling and event-driven control mechanisms. It also includes a pedestrian request feature to ensure safe and efficient crossing. Furthermore, the system integrates real-time monitoring through UART communication and a Python-based interface, which provides live updates and data logging. A prediction model is also incorporated to analyze traffic patterns and improve signal efficiency. Overall, the proposed system aims to reduce congestion, improve traffic flow, enhance safety, and contribute to the development of smart city infrastructure.

II. RELATED WORKS

Several research studies have been conducted to enhance traffic management systems using intelligent and adaptive approaches. Early works primarily focused on sensor-based traffic control systems, where vehicle density is detected using infrared, proximity, or ultrasonic sensors, and signal timings are adjusted accordingly to reduce waiting time and improve traffic flow; however, these systems are often affected by environmental conditions and lack scalability for complex intersections. Microcontroller-based intelligent traffic systems were later developed as cost-effective solutions that prioritize roads with higher traffic density, but they do not support advanced features such as pedestrian prioritization, interconnection between multiple intersections, or real-time monitoring. With technological advancements, IoT-based traffic systems emerged, enabling real-time data collection, remote monitoring, and dynamic signal control through cloud

platforms, though they introduce challenges such as dependency on internet connectivity, increased system complexity, and security concerns. More recent approaches utilize image processing and machine learning techniques for accurate vehicle and pedestrian detection and traffic prediction, but these methods require high computational resources, are sensitive to environmental conditions, and involve higher implementation costs. Additionally, wireless sensor networks and multi-agent systems have been proposed to enable communication between intersections for large-scale traffic optimization, but they face issues related to network reliability, energy consumption, and infrastructure complexity. From the analysis of existing works, it is evident that there is a need for a cost-effective, real-time, and scalable solution that integrates traffic density control, pedestrian safety, monitoring, and predictive analysis, which is effectively addressed by the proposed smart traffic signal controller system.

III. METHODOLOGY

The development of the Smart Traffic Signal Controller using STM32 follows a systematic and structured methodology to ensure efficient traffic management, real-time operation, and reliable performance. The system integrates embedded hardware, real-time operating systems, communication interfaces, and data analysis techniques to achieve adaptive traffic control at a four-way intersection. The methodology includes system design, traffic control logic implementation, event handling, data communication, monitoring, and performance evaluation.

A. System Design and Architecture

The system is designed using an STM32 microcontroller integrated with FreeRTOS to enable multitasking and real-time control. The architecture supports a four-way traffic intersection, where each direction is controlled through a structured signal sequence. The design ensures efficient task scheduling, synchronization, and responsiveness to dynamic traffic conditions.

B. Traffic Control Algorithm

A time-based scheduling algorithm is implemented to control the sequence of traffic signals. Each lane is allocated a predefined time slot for smooth and organized traffic flow. The algorithm ensures fair distribution of signal timing while maintaining system stability and predictability.

C. Event-Driven Signal Control

To improve adaptability, an event-driven mechanism is incorporated into the system. Traffic density and priority requests are used to dynamically adjust signal timings. When higher traffic density is detected on a particular road, the green signal duration is extended. Additionally, pedestrian requests are given priority by temporarily overriding the normal signal sequence to ensure safe crossing.

D. Real-Time Data Communication

UART communication is used to transmit real-time traffic data from the STM32 microcontroller to an external system. This enables continuous monitoring and analysis of traffic conditions. The communication interface ensures reliable and efficient data transfer for further processing.

E. Monitoring and Data Logging

A Python-based monitoring system is developed to visualize and store traffic data. The system logs real-time data into Excel files and provides live updates through Telegram notifications. This allows remote monitoring and performance evaluation of the traffic system.

F. Predictive Analysis and Optimization

A prediction model is implemented to analyze traffic patterns by comparing actual and expected signal usage. The system identifies imbalances in traffic flow and suggests optimized signal timings. This data-driven approach enhances system efficiency and supports adaptive traffic control.

G. System Integration and Testing

All modules, including traffic control logic, event handling, communication, and monitoring, are integrated into a unified system. The system is tested under different traffic scenarios to evaluate performance, responsiveness, and reliability. The results demonstrate improved traffic flow, reduced waiting time, and enhanced pedestrian safety.

H. Design Considerations

Scalability: The system can be extended to multiple intersections with minimal modifications. **Reliability:** Real-time control and event handling ensure consistent system performance. **Cost Efficiency:** The use of STM32 and embedded design reduces overall implementation cost. **Practical Applicability:** The system is suitable for smart city traffic management and real-world deployment.

IV. HARDWARE ARCHITECTURE

The hardware architecture of the Smart Traffic Signal Controller is centered around the STM32 Microcontroller, which functions as the main control unit responsible for processing inputs, executing control algorithms, and managing real-time traffic operations. The system incorporates traffic density sensors such as IR or ultrasonic sensors placed at each road of a four-way intersection to detect vehicle presence and provide real-time input to the controller for adaptive signal timing. A pedestrian push button module is integrated to allow safe crossing requests, which are handled through interrupt-based control by temporarily overriding the normal traffic sequence. The output section consists of traffic signal LEDs (red, yellow, and green) for each direction, controlled through the microcontroller's GPIO pins to regulate vehicle movement. Additionally, a UART communication interface is used to transmit real-time traffic data to an external system for monitoring and analysis. A regulated power supply unit

ensures stable operation of all components, while the complete setup is implemented using a breadboard or PCB with proper interconnections. Overall, the hardware architecture integrates sensing, processing, communication, and actuation modules into a unified system, enabling efficient, real-time, and adaptive traffic management.



Fig. 1. Hardware setup

V. SOFTWARE DESCRIPTION

The developed system provides a comprehensive real-time monitoring, logging, and notification mechanism through an integrated software framework that ensures both immediate visibility and long-term analysis of traffic operations. Traffic data generated by the STM32 Microcontroller is transmitted continuously via UART communication to a Python-based application, where it is parsed, processed, and structured for further use. The system automatically records detailed traffic parameters into an Excel file, including signal state transitions such as NS GREEN, EW GREEN, and BOTH YELLOW, along with precise start time, end time, duration, and priority classification (normal or extended). This structured logging enables chronological tracking of signal behavior and helps in analyzing traffic flow patterns, signal efficiency, and frequency of priority-based extensions. The use of timestamped entries ensures accurate performance evaluation and facilitates identification of peak traffic periods and imbalances between different directions. In addition to offline data storage, the system provides real-time user interaction through a Telegram-based notification interface, where instant updates about traffic conditions are displayed, including current signal status, extended durations, and priority directions such as EAST-WEST or NORTH-SOUTH. This allows remote monitoring of the system without the need for direct hardware access. Furthermore, the software incorporates a short-term prediction module that processes recent traffic data (e.g., 1-minute intervals) to calculate total signal usage, count extension occurrences,

and evaluate traffic dominance in each direction. Based on this analysis, the system intelligently suggests whether signal timings should remain normal or require adjustment, thereby supporting adaptive decision-making. The integration of real-time alerts, automated Excel logging, and predictive analytics creates a robust and scalable software ecosystem that improves transparency, enhances traffic optimization, and enables data-driven traffic management suitable for smart city applications. The system is perfect, for monitoring and can be used in many other applications.

VI. REAL-TIME MONITORING AND DATA LOGGING USING TELEGRAM AND EXCEL

To further enhance system accessibility, monitoring, and intelligent analysis, the proposed smart traffic signal controller incorporates an integrated framework combining real-time communication, predictive analytics, and structured data management using Telegram and Excel platforms. The STM32 microcontroller continuously acquires traffic-related parameters such as vehicle density, signal phase transitions, timing intervals, and priority conditions, and transmits this data through UART communication to a Python-based processing unit. This Python application functions as a centralized middleware that efficiently decodes, processes, and distributes the incoming data simultaneously to multiple interfaces for visualization and storage. For real-time monitoring, the system leverages a Telegram bot that delivers instant, time-stamped notifications containing detailed information such as current signal states (NS GREEN, EW GREEN, BOTH YELLOW), active durations, cycle transitions, and dynamic priority adjustments triggered by congestion or emergency scenarios. This ensures seamless remote accessibility and allows users to track system behavior without physical interaction. Additionally, the system enhances intelligence through a short-term predictive module that analyzes recent traffic trends, identifies imbalances in directional flow, and suggests adaptive signal extensions to optimize throughput and minimize waiting time, thereby improving overall traffic efficiency. Parallel to real-time communication, the system implements an automated data logging mechanism using Excel through Python libraries, ensuring comprehensive storage of traffic events in a structured and chronological format. Each logged entry includes parameters such as signal phase, precise start and end timestamps, duration, vehicle density estimation, and classification of operation mode (normal or priority-based extension). This organized dataset enables in-depth analysis of traffic patterns, detection of peak congestion hours, evaluation of signal performance, and identification of inefficiencies in the control strategy. Furthermore, the stored data can be utilized for statistical analysis, graphical visualization, and future model training for intelligent traffic prediction systems. The dual integration of Telegram for instant alerts and Excel for persistent storage creates a robust monitoring architecture that supports both short-term responsiveness and long-term analytical insights. This approach significantly improves transparency, reliability, and decision-making capabilities while also ensuring scalability for

smart city applications. Moreover, the system design allows future enhancements such as cloud database integration for centralized monitoring, incorporation of machine learning algorithms for adaptive signal optimization, real-time dashboard visualization, and integration with IoT sensors, cameras, and emergency response systems, thereby transforming the solution into a fully autonomous and intelligent traffic management framework.

State	Start	End	Duration	Type
BOTH YELLOW	13:04:44	13:04:45	1.0	NORMAL
NS GREEN	13:04:45	13:04:47	1.47	NORMAL
BOTH YELLOW	13:04:47	13:04:49	2.0	NORMAL
EW GREEN	13:04:49	13:04:50	1.0	NORMAL
BOTH YELLOW	13:04:50	13:04:52	1.92	NORMAL
NS GREEN	13:04:52	13:04:54	2.0	NORMAL
BOTH YELLOW	13:04:54	13:04:55	1.0	NORMAL
EW GREEN	13:04:55	13:04:56	1.0	NORMAL
BOTH YELLOW	13:04:56	13:04:57	1.0	NORMAL
NS GREEN	13:04:57	13:04:59	1.80	NORMAL
BOTH YELLOW	13:04:59	13:05:00	1.0	NORMAL
EW GREEN	13:05:00	13:05:01	1.0	NORMAL
BOTH YELLOW	13:05:01	13:05:04	3.0	NORMAL
NS GREEN	13:05:04	13:05:05	1.0	NORMAL
BOTH YELLOW	13:05:05	13:05:06	1.0	NORMAL
EW GREEN	13:05:06	13:05:07	1.0	NORMAL
BOTH YELLOW	13:05:07	13:05:11	4.0	PRIORITY
NS GREEN	13:05:11	13:05:14	3.0	NORMAL
BOTH YELLOW	13:05:14	13:05:15	1.0	NORMAL
NS GREEN	13:05:15	13:05:17	2.0	NORMAL
BOTH YELLOW	13:05:17	13:05:18	1.0	NORMAL
EW GREEN	13:05:18	13:05:21	3.0	NORMAL
BOTH YELLOW	13:05:21	13:05:25	4.0	NORMAL
NS GREEN	13:05:25	13:05:29	4.0	NORMAL

Fig. 2. Real-Time Traffic Data Logging in Excel

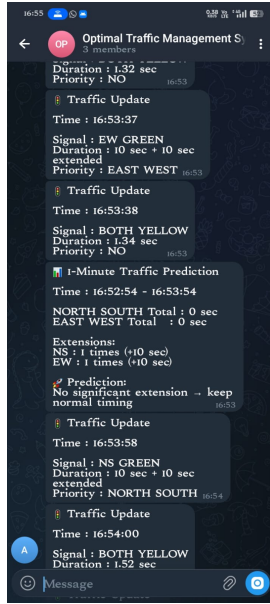


Fig. 3. Real-Time Traffic Updates via Telegram Interface

VII. RESULTS

The proposed Smart Traffic Signal Controller system was successfully implemented and evaluated under different traffic scenarios, demonstrating efficient real-time operation and adaptability. The system dynamically adjusted signal timings based on traffic density using a combination of time-based scheduling and event-driven control, effectively reducing unnecessary waiting time and improving overall traffic flow. Priority-based signal extensions were triggered when higher traffic density was detected, ensuring better utilization of green signal time and minimizing congestion. The pedestrian request feature operated reliably by safely interrupting the normal

signal sequence and allowing controlled crossing. The integration of FreeRTOS enabled smooth multitasking, ensuring that signal control, sensor processing, communication, and data logging were executed without delay. Real-time monitoring through the Telegram interface provided instant updates on signal states, durations, and priority conditions, enabling remote supervision of the system. Additionally, the Excel-based logging system recorded all traffic events with accurate timestamps, allowing detailed analysis of traffic patterns and system performance. The prediction module further enhanced the system by analyzing recent traffic data to identify trends, evaluate extension frequency, and suggest whether signal timings should remain normal or be adjusted. Overall, the results demonstrate that the proposed system significantly improves traffic efficiency, reduces congestion, enhances safety, and provides a cost-effective and scalable solution suitable for smart city traffic management applications.

VIII. CONCLUSION

The proposed Smart Traffic Signal Controller using an STM32 microcontroller effectively overcomes the limitations of conventional fixed-time traffic systems by introducing an adaptive, intelligent, and real-time traffic management approach. The system dynamically adjusts signal timings based on real-time traffic density and incorporates an event-driven mechanism to handle priority conditions such as high vehicle load and pedestrian crossing requests, thereby improving traffic flow, reducing waiting time, and enhancing overall road safety. The use of FreeRTOS ensures efficient multitasking and reliable system performance, allowing simultaneous execution of traffic control, sensor processing, communication, and monitoring tasks without latency. The integration of UART-based communication with a Python interface enables continuous real-time data transmission, while the implementation of Telegram-based notifications provides instant remote monitoring of traffic conditions. Furthermore, automated data logging in Excel facilitates structured storage and detailed analysis of traffic patterns, helping in performance evaluation and system optimization. The inclusion of a prediction module enhances the intelligence of the system by analyzing recent traffic trends, identifying imbalances, and suggesting optimized signal timing strategies. The system is designed to be cost-effective, scalable, and easy to implement, making it suitable for deployment in urban environments and smart city infrastructures. In addition, its modular architecture allows future enhancements such as IoT integration, multi-intersection coordination, emergency vehicle prioritization, and AI-based traffic prediction. Overall, the project demonstrates a practical and efficient solution for modern traffic management, contributing to reduced congestion, improved fuel efficiency, lower environmental impact, and smarter transportation systems.

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